



GPR56 signaling pathway network and its dynamics in the mesenchymal transition of glioblastoma

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Abstract

G protein-coupled receptor 56 (GPR56/ADGRG1) is a multifunctional adhesion GPCR involved in diverse biological processes ranging from development to cancer. In our earlier study, we reported that GPR56 is expressed heterogeneously in glioblastoma (GBM) and is involved in the mesenchymal transition, making it a promising therapeutic target (Ganesh et al., 2022). Despite its important role in cancer, its mechanism of action or signaling is not completely understood. Thus, based on transcriptomic, proteomic, and phosphoproteomic differential expression data of GPR56 knockdown U373-GBM cells included in our above study along with detailed literature mining of the molecular events plausibly associated with GPR56 activity, we have constructed a signaling pathway map of GPR56 as may be applicable in mesenchymal transition in GBM. The map incorporates more than 100 molecular entities including kinases, receptors, ion channels, and others associated with Wnt, integrin, calcium signaling, growth factors, and inflammation signaling pathways. We also considered intracellular and extracellular factors that may influence the activity of the pathway entities. Here we present a curated signaling map of GPR56 in the context of GBM and discuss the relevance and plausible cross-connectivity across different axes attributable to GPR56 function.

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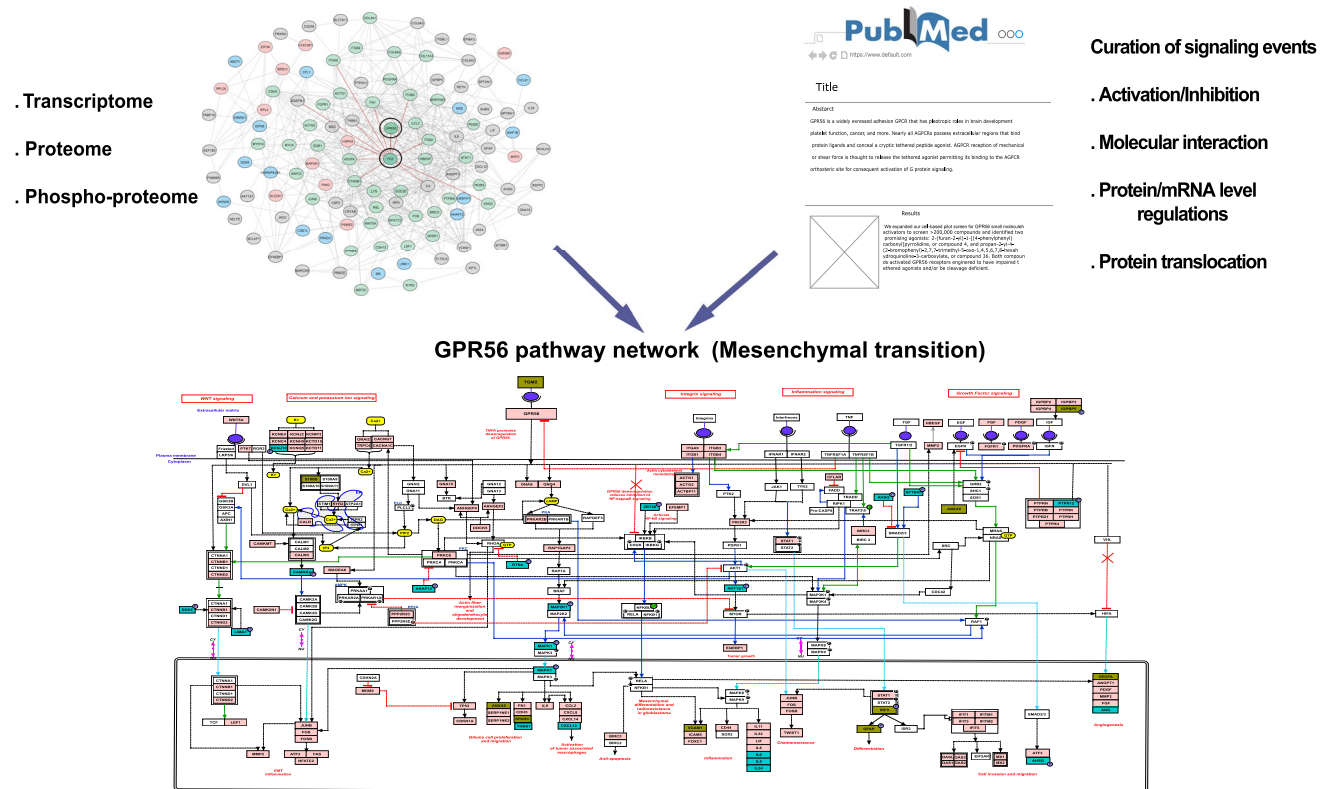
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Graphical abstract

GPR56 signaling and mesenchymal transition

GPR56 knockdown GBM cells

Literature knowledge



Keywords GPR56 /ADGRG1 · TG2 · Signaling · Pathway · Mesenchymal transition

Abbreviations

aGPCRs	Adhesion G protein-coupled receptors
Ca ²⁺	Calcium
CL	Classical
CTF	C-terminal fragment
ECM	Extracellular matrix
ECM	Extracellular matrix
EMT	Epithelial to mesenchymal transition
EMT	Epithelial-mesenchymal transition
GAIN	GPCR auto-proteolysis inducing domain
GBM	Glioblastoma
GDP	Guanosine diphosphate
GPR56	G-protein-coupled receptor 56
IP3Rs	Inositol 1,4,5-trisphosphate receptors
MES	Mesenchymal subtype
MVP	Microvascular proliferation
N	Neural
NF-κB	Nuclear factor kappa B
NTF	N-terminal fragment
PAN	Pseudo-palisading cells around necrosis

PLL	Pentraxin/laminin/neurexin sex-hormone-binding globulin-like
PN	Proneural subtype
TAMs	Tumor-associated macrophages

Introduction

Adhesion G protein-coupled receptors (aGPCRs) are a distinct subgroup within the GPCR superfamily that mediate essential cell–cell and cell–matrix interactions, influencing various physiological processes from embryonic development to immune responses. Their distinctive structure, coupling mechanisms, and intricate signaling pathways make aGPCRs intriguing targets for studying fundamental cellular processes and investigating potential therapeutic strategies (Langenhan et al. 2016; Purcell and Hall 2018). G protein-coupled receptor 56 (GPR56/ADGRG1), an aGPCR plays a critical role in developmental processes such as cortical development and neuronal migration as well as a significant

role in cancer pathogenesis by modulating tumor growth, invasion, and metastasis (Ganesh et al. 2020). The human GPR56 gene is located on chromosome 16q21 and has 14 exons spanning 15 kb of genomic sequence and a 3-kb open reading frame (Ke et al., 2008; J. E. Kim et al., 2010; Piao et al., 2004). GPR56, like other adhesion GPCRs, has a large extracellular N-terminus, followed by seven transmembrane alpha helices joined by three interhelical loops on both sides of the plasma membrane and an internal C-terminus (Araç et al. 2012). The extracellular region consists of a PLL (pentraxin/laminin/neurexin sex-hormone-binding globulin-like) domain and a GPCR auto-proteolysis inducing (GAIN) domain (Salzman et al. 2016). GAIN domain mediates auto-proteolytic cleavage in response to ligand binding, resulting in an N-terminal fragment (NTF) and a C-terminal fragment (CTF) that initiates downstream signaling processes (Salzman et al. 2016, 2017). GPR56 ligands include ECM components/proteins such as collagen-III and transglutaminase-2 (TG2). Both of these proteins have been reported to bind to the PLL domain and facilitate NTF dissociation after the autoproteolytic cleavage to initiate downstream signaling events through the G protein activation cascade (Luo et al. 2012; Yang et al. 2014).

Altered expression of GPR56 has been observed in many cancers including brain tumors like Glioblastoma (GBM) (Ackerman et al. 2015; Ke et al. 2007; Piao et al. 2004; Shashidhar et al. 2005). In fact, GPR56 is one of the key aGPCRs involved in GBM (Stephan et al. 2021). GBM is a highly aggressive brain tumor characterized by its complex molecular and genetic heterogeneity, which has led to the identification of distinct molecular subtypes- (neural (N), proneural (PN), classical (CL), and mesenchymal (MES)) (Patel et al. 2014; Verhaak et al. 2010; Wang et al. 2017). In our earlier publication, we have shown that the GPR56 expression in GBM is heterogeneous—high in PN and low in the MES subtype, and its expression is negatively associated with TG2 in PN and MES cells (Ganesh et al. 2022). The inverse relationship between GPR56 and TG2 was also supported by spatial expression data (microvascular proliferation and pseudopalisading cells around necrosis (PAN) of GBM tissues and also in GBM cells cultured under hypoxic conditions (Ganesh et al. 2022). GPR56 is downregulated by hypoxia in several malignancies (Elvidge et al. 2006; Maina et al. 2005), whereas TG2 is elevated in response to hypoxia (Eckert 2019). According to Yang et al. (2014), TG2 binding to GPR56 promotes its internalization and degradation in melanoma, therefore it is possible that the same mechanism may hold true for the reciprocal expression of TG2 and GPR56 in GBM although other modes of regulation such as transcriptional changes cannot be ruled out. Given that TG2 is involved in crosslinking ECM proteins and ECM remodeling and is elevated in mesenchymal GBM (Yin et al. 2017), we believe that the interaction between GPR56 and TG2

and their reciprocal expression may be associated with the mesenchymal transition. Further, hypoxia is also reported to promote mesenchymal shift in GBM by inducing hypoxia-inducible factors (Joseph et al. 2015). Therefore, integrating the connections between these conditions and the underlying molecular events to curate the GPR56 pathway map is pivotal. Thus, we integrated transcriptome data (RNA-seq), differential proteomic and phosphoproteomic data from GPR56 silenced GBM cell line U373 combined with GPR56 signalling related knowledge from the published literature, to curate a pathway map of GPR56 which may be consistent with mesenchymal transition in GBM.

Methods

Annotation of GPR56 signaling events in glioblastoma

The annotation of GPR56 is divided into three parts:

Annotation using multi-omics data from the GPR56 knockdown GBM cell line U373

The molecular entities used in the pathway network are majorly drawn from the differential expression data (transcriptomics/proteomics and phosphoproteomics) acquired in the GPR56 knockdown experiment with GBM cell line U373 (justification for using knockdown data is discussed below). A total of 119 molecular entities including 97 differentially expressed transcripts, 16 proteins, and 21 phosphoproteins with expression levels larger than 1.5 folds were taken into consideration. This included molecules involved in one or more of the top enriched pathways such as Inflammation by Chemokine and cytokine signaling, Wnt, Integrin, Ca signaling, and angiogenesis. These pathways and their pathway axes are known to be associated with the process of mesenchymal transition in general, some of them were also found to be hub entities in the interaction network (Ganesh et al. 2022) (Supplementary Tables 1A and 2A).

Literature-driven annotation

Further, we carried out an extensive literature search from the years 1999 to 2023 in PubMed and Google Scholar using the key terms, “GPR56” OR “ADGRG1” AND “Signaling” OR “Pathway” to annotate the events relevant to GPR56 stimulation and signaling. 288 Articles were screened for information related to molecular interactions, post-translational modifications, changes in protein localization, as

well as gene expression changes, as per metrics described in Kandasamy et al. (2010) (Supplementary Tables 1B and 2B).

Annotation with global differential expression data of GBM

The majority of molecules incorporated into the network (A and B list) are also found to be differentially expressed in global RNA and/or protein expression data of GBM. Although an indirect support, we extended this corollary, to other earlier recognized and established entities of the pathway axes (prior knowledge) that are not covered in the A and B list and captured them from global differential expression data of GBM because of their relevance in the present context (Supplementary Tables 1C and 2C).

Generation of GPR56 signaling pathway network

The data of diverse molecular reaction events induced by GPR56 gathered from the GPR56 knockdown multi-omics data and by manual curation was then used to generate the GPR56 signaling pathway map using “PathVisio” version 3.3.0), an open-source pathway drawing software. The pathway map is presented in Graphical Pathway Markup Language (GPML) format. GPML format is a custom XML format compatible with pathway visualization and analysis tools such as Cytoscape (Shannon et al. 2003) and PathVisio (Kutmon et al. 2015). (Fig. 1).

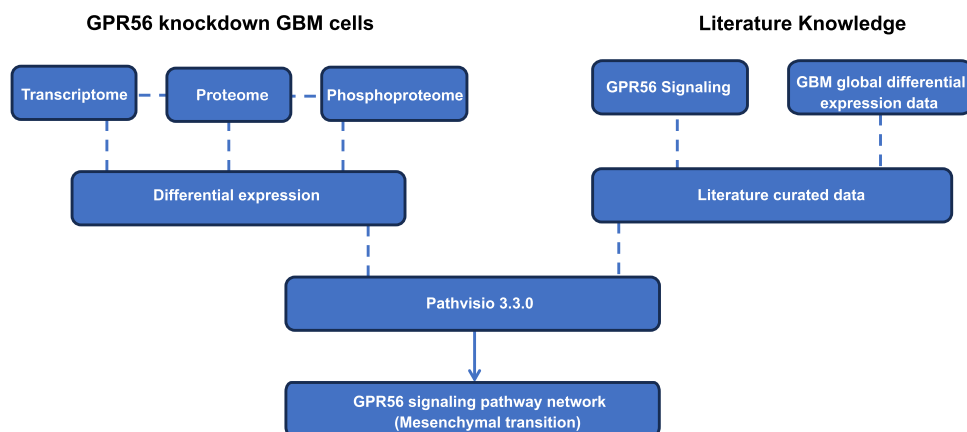
Results and discussion

Generally, signaling pathway information for a receptor is obtained by inducing receptor expression or activity, and pathways are populated by integrating the molecular information thus obtained from pathway stimulation conditions as well as differential gene expression leading to a defined cellular reaction or process. GPR56 is an adhesion GPCR involved in normal development as well as cancer with TG2

being one of the two known protein ligands (Ganesh et al. 2020). Interestingly TG2 itself is a multifunctional protein present intra and extracellularly and interacts even with other cell surface receptors (Nurminskaya and Belkin 2012). Further, the two proteins are also expressed reciprocally in spatial regions of GBM (Ganesh et al. 2022). So, the interaction between GPR56 and TG2 ligand may involve a complex network of molecular cascades connected not only to GPR56 activity but also to the independent activity of TG2, representing an extraordinary regulatory pathway system with implications in the process of mesenchymal transition. Mesenchymal transition being highly dynamic, the process may be a sequential array of morphological as well as molecular steady states comprised of receptor expression and activity changes. Changes in the expression may be regulated independent of the activity state of the receptor with no sharp boundaries but co-occurring in a mutually exclusive manner during the transition phase. Therefore, both knockdown and overexpression or stimulation models of either of the two proteins may yield important insights of this pathway network/system. Thus, for the curation of the pathway network, we have utilized molecular events observed under GPR56 knockdown/depletion in GBM cell line from our earlier study (Ganesh et al. 2022) as detailed below. In addition, we also used molecular information drawn from the literature under GPR56 stimulated as well as knockdown studies. The insights could be verified and confirmed further with additional experimental strategies like over expression of the receptor or knockdown of TG2.

In our earlier study (Ganesh et al. 2022), we generated transcriptome, proteome, and phosphoproteome data for GPR56 knockdown U373 GBM cells. Our analysis showed 1010 transcripts, 47 proteins, and 70 phosphoproteins that were differentially expressed (Ganesh et al. 2022). The proteomic data used here is relatively small on account of the workflow used (although could be expanded using workflows for deeper coverage). Thus, we primarily used the transcriptome data to map the altered pathways in GPR56

Fig. 1 Workflow for identification and integration of molecular profile for incorporation into GPR56 pathway map



knock-down cells, using the PANTHER database and supported it with the proteomic data. Pathway analysis revealed many key pathways like Inflammation by Chemokine and cytokine signaling, Wnt, Integrin, angiogenesis, and others which are known to be associated with the mesenchymal transition or mesenchymal state in general. Although relatively small in coverage, the proteomic and phosphoproteomic data used did corroborate with the transcriptomic data from GPR56 knockdown GBM cells concerning the key entities implied. Further, we also captured relevant information related to GPR56 signaling from published literature as described in the Methodology section. The overall workflow of the study is shown in Fig. 1. Over 288 research articles related to GPR56/ADGRG1 and its signaling were screened. Information on PPIs, enzyme-catalyzed events, protein translocations, site-specific PTMs, and activation/inhibition events in response to the respective ligands was extracted and annotated. Given the potential role of GPR56 signaling in cellular processes such as migration, invasion, and angiogenesis putatively impacting GBM pathology, we kept the following observations in the background namely, reciprocal expression of GPR56 and TG2, their differential interaction in the two cellular states (MES and PN), Calcium (Ca^{2+}) homeostasis and its influence on the activity states of TG2 and the consequent dynamics of TG2, extracellular matrix (ECM), and Integrins. To have an integrated

view of all entities implicated in these processes, we built a curated pathway map using this multi-omics data from GPR56 knockdown cells. The map consists of 200 or more putative interactions altered upon loss of GPR56 which are shown in different colored boxes (Fig. 2), based on whether they are detected at transcriptomic and proteomic levels or both. We discuss the putative role of the various pathway axes and interactions that may be attributed to GPR56 signaling in GBM.

Wnt pathway is known to be a major pathway connected to invasiveness and metastasis in many cancers and is also reported to be associated with the expression and activation of genes associated with epithelial-mesenchymal transition (EMT) in glioma cells (Lee et al. 2016). Along with its co-receptors, the Wnt family of glycoproteins bind to the extracellular domain of Frizzled receptors extracellular domain, activating the Wnt signaling. Activation of Wnt signaling leads to phosphorylation of β -catenin via GSK-3 activation and Axin displacement, resulting in its nuclear translocation and eventual interaction with transcriptional activators TCF/LEF (Lee et al. 2016). Nuclear translocation of β -catenin is further aided by its association with phosphorylated DDX5, which also co-activates transcription factors and matrix metalloproteases like MMP2 involved in EMT and proliferation. (Nyamato et al. 2019). In GPR56 knockdown data, Wnt5A, its coreceptor

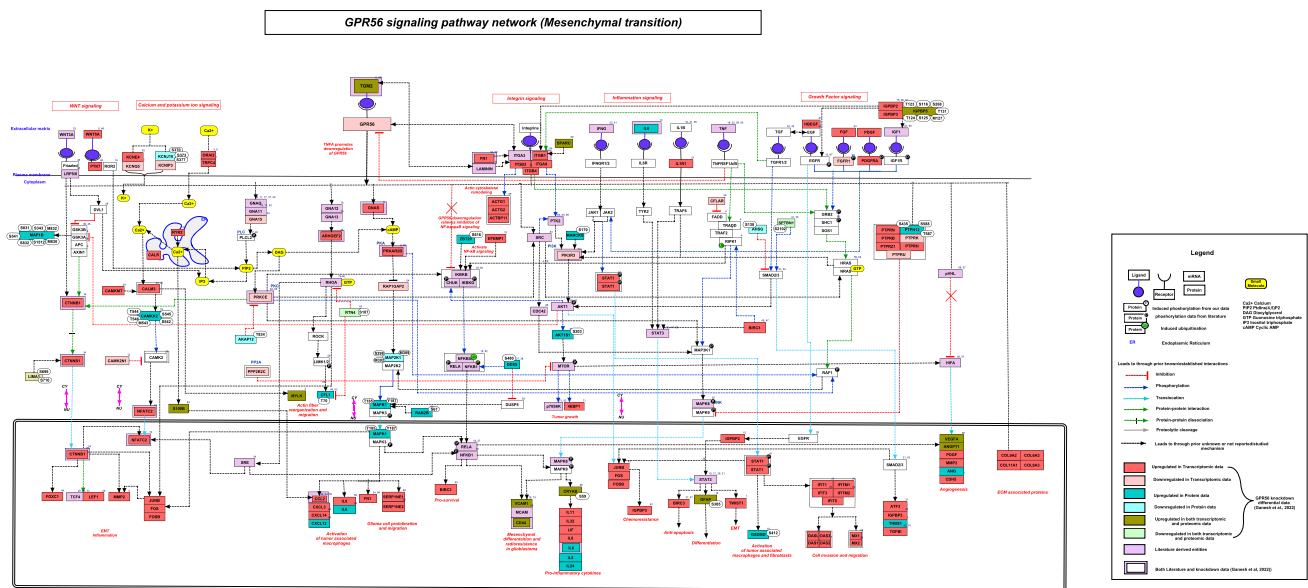


Fig. 2 A schematic pathway network of molecular interactions involved in GPR56 signaling applicable in the context of mesenchymal transition of GBM. The pathway reaction map depicts various types of reactions largely on expression data driven information induced upon loss of GPR56 in GBM cells and include changes such as molecular interactions, catalysis, and translocation events. Relevant knowledge from published literature is also drawn to complement data driven reactions. GPR56 knockdown driven

expression changes - up or downregulation at transcript or protein level, and literature driven gene entities are distinguished by colors and color codes are provided. The site and residue information of post-translational (phosphorylation) modifications are provided for GPR56 knockdown phosphorylation data. Different types of reactions in the pathway map are distinguished with different colors and arrows as provided in the legend. (High-resolution figure is attached separately)

PTK7, as well as several β -catenin's (CTNNB1, CTNND2, CTNNA2) the major mediators of Wnt signaling were found to be upregulated. Upregulation of Wnt pathway components in GBM, specifically β -catenin, suggests a role in both invasion and proliferation through different pathway nodes (Nager et al., 2012). GPR56 signaling has been shown to modulate the Wnt pathway in GBM and adipogenesis (Al Hasan et al. 2020; Shashidhar et al. 2005). Besides that, TG2 also plays an important role in the stabilization and nuclear translocation of β -catenin by binding to LRP5/6, inducing the TCF /LEF transcription factors, and decreasing CDKN1A expression (Condello et al. 2022). TG2 inhibition has also been shown to suppress Wnt signaling by reducing β -catenin levels (Han et al. 2020). Thus, these molecular changes observed in the absence of GPR56 and the increased expression of TG2 support the cross talk/connectivity between GPR56 and Wnt signaling.

Integrins play a significant role in many physiological and pathological conditions including GBM by linking the intracellular cytoskeleton to the extracellular environment (Bianconi et al. 2016). Several ECM proteins, including fibronectin, collagen, osteonectin, thrombospondins, and enzymes like MMPs interact with integrins to modulate cell–matrix interactions by disrupting focal adhesions and reorganizing actin stress fibers (Bornstein 2009, Cawston and Young 2010). In GBM, integrins promote mesenchymal transition by recruiting focal adhesion kinase (FAK) which controls the cytoskeletal organization, and survival, or by activating additional pathways (Shc/MAP-Kinases, Rho-GTPases, Src Family Kinases) (Malric et al. 2017). Several integrins (ITGA9, ITGB1, ITGB3, ITGB4), ECM proteins, and enzymes (FN1, MMP-2, MMP-17, SPARC, THBS1, and ADAMTS7/9/SL1) were found to be overexpressed in the glioma mesenchymal cells were found to be upregulated in GPR56-Knockdown cells (Malric et al. 2017). Studies have shown that GPR56 shares a synergistic relation with $\alpha 3 \beta 1$ -integrin in developing brain to regulate cell migration (Jeong et al. 2013) and it has been shown that GPR56 inhibits fibronectin deposition and its downstream signaling, such as phosphorylation of FAK (Millar et al. 2018). Also, the N-terminal domain of TG2 is known to bind to fibronectin extremely well, facilitating the formation of a complex with integrins, essential for maintaining cell adhesion in the matrix (Akimov et al. 2000; Belkin 2011). TG2-induced integrin clustering potentiates integrin-dependent intracellular signaling that modulates activities of FAK, Src, and p190RhoGAP and increased expression of RhoA GTPase and its downstream target, ROCK (Nurminskaya and Belkin 2012). The net impact of these events is increased focal adhesion and actin stress fiber formation leading to enhanced actomyosin contractility (Eckert et al. 2014). Additionally, TG2/Fibronectin integrin complex formation on the surface

of cancer cell membranes was shown to regulate β -catenin expression and loss of TG2 also decreased mesenchymal markers like MMP-2, TWIST-, CD44, etc. (Condello et al. 2022; Sima et al. 2022), thus implying a cross-talk between GPR56—TG2 dynamics and integrin signaling associated with the cytoskeletal reorganization and cell invasion.

Pro-inflammatory cytokines, chemokines, and growth factors majorly enrich the tumor microenvironment in GBM and influence tumor progression through complicated cross-talk with their receptors (Motaln and Turnsek 2015; Shi et al. 2017). GBM cells also utilize chemokine gradients to draw microglia and monocytes, which leads to the accumulation of tumor-associated macrophages (TAMs) to create an immunosuppressive microenvironment (Susek et al. 2018). We observed that several chemokines and cytokines including IL6, IL8, IL5, IL24, CXCL12, CCL2, CXCL8, and CXCL14 were found to be upregulated in GPR56-deficient GBM cells. Negative regulation of GPR56 by cytokines was also shown in GBM by Moreno et al. (Moreno et al., 2017) where cytokine TNF- α promotes mesenchymal transition by suppressing GPR56 expression and reversing GPR56 mediated NF- κ B inhibition. While GPR56 is shown to negatively regulate NF- κ B signaling, TG2 activates NF- κ B signaling. On the other hand, TG2 itself is activated through TNF- α mediated NF- κ B signaling thereby forming a feedback loop. Another pro-inflammatory cytokine—TGF- β 1, promotes the expression of TG2, which in turn binds to the cell surface and converts the cytokine to its biologically active form, resulting in a positive feedback loop (Quan et al. 2005). TG2 expression can also be enhanced by IL-6, IL-8, and IL-1 (Eckert 2019). Thus, the combined effect of the cytokines released may induce TG2 expression intracellularly followed by its extracellular translocation under the deregulated Ca^{2+} flux (see below). Furthermore, while GPR56 expression is low in TAMs, TG2 is abundant (Afik et al. 2016; Bilemjian et al. 2022). It appears that TAMs are involved in tumor development by remodeling the ECM structure and composition by covalently cross-linking tumor-associated molecular signatures to TG2, thereby stabilizing and improving the ECM structure and composition (Afik et al. 2016). Therefore, the combined effect of the cytokines, chemokines, and TAMs may stimulate TG2 expression and dynamics. Extracellular TG2 then interacts with growth factor receptors and may activate downstream signaling (Belkin 2011). Many growth factors, such as FGF, PDGF, VEGF, or IGF, stimulate receptor tyrosine kinases (RTKs), which activate various signaling cascades, including the Ras, PI3K, Src, and FAK pathways (Jechlinger et al. 2006; Kim et al. 2007; Murillo et al. 2005; Xie et al. 2020). Physical interaction between integrins and receptor tyrosine kinases is required for cell response to ECM and soluble growth factors (Eckert et al.

2014). For example, TG2 interacts with PDGFR on the cell surface and improves PDGFR interaction with integrins by bridging these receptors. PDGFR clustering, PDGF and adhesion-induced PDGFR activation, and downstream signaling are all aided by the interaction with TG2 (Zemskov et al. 2009). Also, TG2 has been found to interact with VEGFR on the surface of endothelial cells to modulate VEGF-mediated signaling (Dardik and Inbal 2006), while GPR56 negatively regulates VEGF production by interacting with TG2 (Yang et al. 2011). All these molecular events may be interconnected in the overall pathway network.

Ion channels are significantly altered in GBM, changes in membrane affect pH, nutrient transport, and Ca^{2+} homeostasis has been linked to proliferation, migration, and invasion (Leclerc et al. 2016; Polisetty et al. 2012). Several Ca^{2+} permeable ions channels and transporters regulate Ca^{2+} concentration in the cytosol. The major source of Ca^{2+} increase is the release of intracellular Ca^{2+} reserves via Inositol 1,4,5-trisphosphate receptors (IP3Rs). Several RTK and GPCR agonists activate IP3Rs in GBM cells, increasing Ca^{2+} release from the ER (Kang et al. 2010). Additionally, it is known that Ca^{2+} regulates the levels of protein kinase, downstream calmodulin-dependent (CAM) kinase pathways, and activation of several transcription factors in GBM (Chigurupati et al. 2010; Leclerc et al. 2016). Several calcium regulatory channel proteins, including TRPC4, ORAI2, and RYR2, as well as sensors like S100B, were found to be upregulated in our GPR56-deficient knockdown GBM cells (Ganesh et al. 2022). While GPR56 is implicated in regulating intracellular Ca^{2+} concentration in glioma (Ohta et al. 2015), TG2 has been shown to regulate IP3R-dependent Ca^{2+} influx through ligand-gated channels (Hamada et al. 2014). Furthermore, we also found that calmodulin (CALM), CAMKK2, and transcription factor NFATc2 were all upregulated in our GPR56 knockdown cells implying activation/changes in the Ca^{2+} signaling pathway.

Taken together, there appears to be a functional link between hypoxia, mesenchymal transition, and GPR56 TG2 dynamics with subsequent effects on Ca^{2+} homeostasis, and regulation of ion channel, interactions with ECM and other cell surface receptors, all of which form a complex network of signaling cascades. The existence of the GPR56 splice form 4 lacking NTF with ligand binding site raises an interesting possibility of NTF dynamics and its accessibility for TG2 interaction in the mesenchymal state, although experimental support is currently lacking. Regardless, Fig. 2 incorporates key entities differentially altered in GPR56 knockdown cells along with information from published literature supporting the above putative functional links in the curated GPR56 pathway that is contextually aligned with the process of mesenchymal transition in GBM. The pathway network presented here is thus populated more

with data-driven information from the receptor knockdown experiments as lowering of the receptor was found to align with mesenchymal transition, a highly complex process in cancer progression. Although consistent from the biological perspective, the specific leads incorporated in the pathway map and the therapeutic cues maybe further investigated through targeted cell and activity-based experiments.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s12079-023-00792-5>.

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Data availability Data generated during this study are available at GitHub via following URLs: <https://github.com/Raksha-ganesh/GPR56-pathwaymap/blob/main/GPR56%20pathway%20figure%20pdf.pdf>, <https://github.com/Raksha-ganesh/GPR56-pathway-map/blob/main/GPR56%20pathway%20figure.gpml>

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